

Configuring Firewalls Using IPTables



Andrew Mallett

Author and Trainer

@theurbanpenguin www.theurbanpenguin.com



Overview



Firewalling Using IPTables

- Network ports
 - udp
 - tcp
 - /etc/services
 - nmap
 - --top-port
- Firewalls and iptables
- Transient and permanent rules



Open Ports: Having ports open on your system is the same as having doors and windows in a building. These act as entry points for both welcome and unwelcome visitors



```
$ ss -ntl  
$ ss -nul  
$ less /etc/services  
$ sudo apt install -y nmap  
$ nmap -v -oG -  
$ nmap --top-port 10 -oG -
```

Listening Ports

Connecting to a service across the network requires the service to be started and listening on a port. The port can be managed via the TCP or UDP transport layer protocol. A good starting audit on any system is to list the TCP and UDP ports that are open. Installing a port scanning tool such as nmap can be useful to list common ports found on internet systems

Demo



Understanding Attack Vectors:

- Ubuntu 1
 - list open ports
 - view ports in /etc/services
 - use nmap to display top ports



Implementing a firewall allows
you to control access to a
network or individual host



```
$ sudo iptables -L
$ sudo apt list '*.iptables*'
$ sudo apt install -y iptables.persistent
$ systemctl cat iptables.service
$ file /usr/sbin/netfilter-persistent
$ cat /usr/sbin/netfilter-persistent
$ cat /etc/default/netfilter-persistent
$ cat /etc/iptables/rules.v4
```

Firewall Rules Using Netfilter Kernel Modules and Iptables

The command `iptables` can be used to manage the host based netfilter firewall. These are kernel modules that are present by default but without active rules. We can install additional packages to allow ease of access to systemd service unit to load rules on boot

```
$ who
$ sudo iptables -A INPUT -p tcp -s 10.0.2.2 --dport 22 -j ACCEPT
$ sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
$ sudo iptables -A INPUT -i lo -j ACCEPT
$ sudo iptables -A INPUT -j DROP
$ sudo iptables -L
```

Setting Transient Rules

Firewall rules reside in memory. For the moment, we only want allow our client IP address to SSH. We can check the IP used from who. Additionally, we need to allow established and related connections as often we connect to different ports once a connection is made, we also need to allow access from the localhost to all services. Rules are read in order with the DROP last.


```
$ sudo iptables -Z && sudo iptables-save -f /etc/iptables/rules.v4
$ cat /etc/iptables/rules.v4
$ sudo iptables -F
$ sudo iptables-restore /etc/iptables/rules.v4
$ sudo iptables -L
```

Setting Persistent Rules

Firewall rules we want to persist should be stored in the file `/etc/iptables/rules.v4` to be accessed by the service unit. We can clear the counters before saving ensuring they are not stored in the file. **Do not store firewall rules if they have caused problems in the runtime configuration**

Demo



Working with Rules:

- Ubuntu 1
 - iptables
 - iptables-save
 - iptables-restore
 - reboot



We can test rules from another device. From Ubuntu 2 SSH packet should be dropped. We can see the counter increment



```
$ sudo apt install -y apache2
$ sudo iptables -vL --line-numbers
$ sudo iptables -I INPUT 3 -p tcp -s 192.168.56.0/24 --dport 80 -j ACCEPT
$ sudo iptables -S
$ sudo iptables -D INPUT 3
$ sudo vim /etc/iptables/rules.v4
$ sudo iptables-restore /etc/iptables/rules.v4
```

Adding Rules

Previously, we appended rules to the chain. With the last rules being the DROP rule appending is not an option. We need to insert at a given line number. We can also delete rules. Personally, I find it easier to edit the rule file and restore it.

Demo



Configuring HTTP Access

- Ubuntu 1
- install apache
- add firewall rule allowing http access from subnet
- test from Ubuntu 2



Summary



Network Firewalls

- nmap
- ss
- iptables
- netfilter-persistent
- iptables-save
- iptables-restore



Configuring Firewalls Using UFW

